Early Prediction of Lithium-Ion Battery Cycle Life with Uncertainty Quantification

Goal. Testing a new battery design or fast-charging protocol until the cells fail can take months. This project will predict a cell's cycle life (cycles until capacity falls to 80% of nominal) from only its earliest cycles, with calibrated confidence intervals. This would let engineers screen designs and protocols much faster [1], [2].

Data. The dataset from [1] contains 124 LFP/graphite cells cycled under 72 fast-charging protocols, with lifetimes of about 150–2,300 cycles. It is publicly available from Toyota Research Institute (data.matr.io).

Methods.

1. *Baseline:* Reproduce the elastic net from [1], which uses the variance of the discharge capacity difference curve $\Delta Q_{100-10}(V)$. Add random forest and gradient boosting for comparison.
2. *Deep model:* Train a 1D CNN in PyTorch on raw $\Delta Q(V)$ curves and per-cycle measurements.
3. *Early-prediction study:* Repeat the experiments using only the first 100, 50, 20, and 10 cycles. For shorter windows, the feature is computed between the last available cycle and an earlier reference cycle.
4. *Uncertainty:* Compare Gaussian process regression with deep ensembles [3].

Evaluation. Models will be scored by RMSE and mean absolute percentage error on the original test splits from [1], allowing direct comparison with published results. Uncertainty methods will be scored by prediction-interval coverage and width.

Expected outcomes. The project will deliver a benchmark of classical and deep models, identify the minimum number of cycles needed for a useful prediction, and assess whether the uncertainty estimates are reliable enough to justify ending a test early.

Timeline. Weeks 1–2: data processing and baseline. Weeks 3–4: CNN and early-prediction study. Weeks 5–6: uncertainty methods. Week 7: analysis and report.

References.

[1] K. A. Severson et al., *Nature Energy*, vol. 4, pp. 383–391, 2019.
[2] P. M. Attia et al., *Nature*, vol. 578, pp. 397–402, 2020.
[3] B. Lakshminarayanan et al., *NeurIPS*, 2017.